Algebra 2

Unit 3: Polynomial Functions

Practice Test — Study Copy — Answer Key

Keep in Mind

This is a practice test. It mirrors the real Unit 3 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

1. Degree [B]
2. Leading coefficient [A]
3. End behavior [E]
4. Multiplicity of a zero [C]
5. Turning point [D]
6. Factor Theorem [F]
7. Rational Root Theorem [G]
8. Fundamental Theorem of Algebra [H]

(A) The number multiplying the highest-power term; its sign decides the right-hand arm.

(B) The greatest exponent on the variable once the polynomial is in standard form.

(C) How many times the factor $(x-r)$ appears — odd means the graph crosses at r, even means it touches.

(D) A point where the graph changes from increasing to decreasing, or the reverse.

(E) What the graph does at its far left and far right, as $x \to \pm\infty$.

(F) $(x-r)$ is a factor of $f(x)$ exactly when $f(r)=0$.

(G) Every rational zero of a polynomial with integer coefficients looks like $\pm\frac{p}{q}$, where p divides the constant term and q divides the leading coefficient.

(H) A polynomial of degree $n \geq 1$ has exactly n complex zeros, counted with multiplicity.

Part B — Multiple Choice (12 pts)

Circle the best answer.

B 1. The end behavior of $f(x) = -2x^4 + 3x^2 - 1$ is (A) both arms up **(B) both arms down** ← (C) left down, right up (D) left up, right down

C 2. $(2x-5)^2 =$ (A) $4x^2 - 25$ (B) $4x^2 + 25$ **(C)** $4x^2 - 20x + 25$ ← (D) $4x^2 - 10x + 25$

A 3. Factored completely, $x^3 + 27$ is **(A)** $(x+3)(x^2-3x+9)$ ← (B) $(x+3)(x^2+3x+9)$ (C) $(x+3)^3$ (D) prime

A 4. The remainder when $f(x) = x^3 - 4x^2 + 1$ is divided by $x-2$ is **(A)** -7 ← (B) -23 (C) 7 (D) 0

C 5. Which value is *not* a possible rational zero of $f(x) = 3x^3 + 2x^2 - x + 4$? (A) -2 (B) $\frac{4}{3}$ **(C) $\frac{3}{4}$** ← (D) 1

A 6. A polynomial with *real* coefficients has degree 4 and zeros 2, -1, and $3i$. Its fourth zero must be **(A) $-3i$** ← (B) $3i$ (C) -3 (D) $\frac{1}{3}$

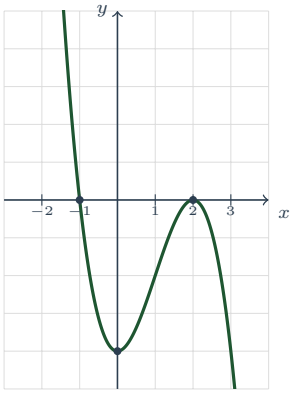
Teacher Note

Item 1: even degree ⇒ the arms match; $a = -2 < 0$ ⇒ they both point down. Only two facts are needed — the $3x^2$ and the -1 are irrelevant. Item 2: (A) and (B) are the $(a-b)^2 = a^2 - b^2$ error, (D) forgets to double the middle product. Item 4: Remainder Theorem, $f(2) = 8 - 16 + 1 = -7$; (B) is $f(-2)$ — the sign-flip error. Item 5: $p \mid 4$ and $q \mid 3$, so the list is $\pm1, \pm2, \pm4, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm\frac{4}{3}$. $\frac{3}{4}$ is the *flipped* fraction. Item 6: real coefficients force conjugate pairs; $3i$ cannot travel alone.

Part C — Short Answer & Computation (40 pts)

Show your work.

1. **Read the graph.** The polynomial function f is graphed below. Each gridline is one unit.



(a) Degree: even or odd? [odd] Leading coefficient: positive or negative? [negative]

(b) As $x \to -\infty$, $f(x) \to$ [$+\infty$]; as $x \to +\infty$, $f(x) \to$ [$-\infty$]

(c) Zeros, each with its multiplicity, and whether the graph crosses or touches there: [$x = -1$ (multiplicity 1, crosses); $x = 2$ (multiplicity 2, touches)]

(d) y-intercept: [(0,-4)]; number of turning points: [2]

(e) Relative maximum: [(2,0)]; relative minimum: [(0,-4)];
absolute maximum or minimum: [none — odd degree, so the arms run to $\pm\infty$]

(f) Interval(s) where f is increasing: [(0,2)]; domain: [all reals]; range: [all reals]

2. **Operations.** Simplify. Write each answer in standard form.

(a) $(3x^3 - 5x + 2) - (x^3 + 4x^2 - 5) = 3x^3 - 5x + 2 - x^3 - 4x^2 + 5 = 2x^3 - 4x^2 - 5x + 7$. **(The subtraction hits *every* term of the second polynomial: -5 becomes $+5$.)**

Answer: [$2x^3 - 4x^2 - 5x + 7$]

(b) $(2x - 3)(x^2 + 4x - 1) = 2x^3 + 8x^2 - 2x - 3x^2 - 12x + 3 = 2x^3 + 5x^2 - 14x + 3$

Answer: [$2x^3 + 5x^2 - 14x + 3$]

(c) For your answer to (b), give the degree [3], the leading coefficient [2], and the end behavior [left down, right up]

3. **Factor completely.** Every factor must be prime over the integers.

(a) $2x^3 - 18x$ (b) $27x^3 - 8$ (c) $x^3 + 3x^2 - 4x - 12$

(a) GCF first: $2x(x^2 - 9) = 2x(x - 3)(x + 3)$.

(b) Difference of cubes with $a = 3x$, $b = 2$: $(3x - 2)\big((3x)^2 + (3x)(2) + 2^2\big) = (3x - 2)(9x^2 + 6x + 4)$**; the quadratic factor is prime.**

(c) Four terms $\Rightarrow$ grouping: $x^2(x + 3) - 4(x + 3) = (x + 3)(x^2 - 4) = (x + 3)(x - 2)(x + 2)$.

(a) $2x(x-3)(x+3)$ **(b)** $(3x-2)(9x^2+6x+4)$ **(c)** $(x+3)(x-2)(x+2)$

4. **Division and the Remainder Theorem.**

(a) Divide using *long division* and write the answer as $Q(x) + \dfrac{R}{D(x)}$: **Answer:** $x^2+2x+6+\frac{3}{x-2}$

$$(x^3 + 2x - 9) \div (x - 2)$$

Write the dividend with a placeholder: $x^3 + 0x^2 + 2x - 9$**. Long division gives** $Q(x) = x^2 + 2x + 6$ **with** $R = 3$**, so the answer is** $x^2 + 2x + 6 + \dfrac{3}{x - 2}$**. Check:** $(x-2)(x^2+2x+6)+3 = x^3+2x-12+3 = x^3+2x-9$ ✓

(b) Use the *Remainder Theorem* to find the remainder when $g(x) = x^3 + 2x^2 - 5x + 6$ is divided by $x + 3$. Is $x + 3$ a factor of g? Explain in one sentence.

$x + 3 = x - (-3)$, so the remainder is $g(-3) = -27 + 18 + 15 + 6 = 12$. Since the remainder is 12 and not 0, $x + 3$ is *not* a factor (the Factor Theorem requires $g(-3) = 0$).

5. **Synthetic division.** You are told that $x = 3$ is a zero of

$$f(x) = x^3 - 2x^2 - 5x + 6.$$

Use synthetic division to divide out that zero, factor f completely, and list *all* of its zeros. **Synthetic division by** $r = 3$ **on** $1,\ -2,\ -5,\ 6$ **gives the bottom row** $1,\ 1,\ -2,\ 0$ **— remainder 0, confirming the zero. The depressed polynomial is** $x^2 + x - 2 = (x + 2)(x - 1)$**, so** $f(x) = (x - 3)(x + 2)(x - 1)$ **and the zeros are** $x = 3,\ x = -2,\ x = 1$**.**

Factored: $(x-3)(x+2)(x-1)$ **Zeros:** $3,\ -2,\ 1$

6. **Rational Root Theorem.** For $f(x) = 2x^3 - x^2 - 8x + 4$:

(a) List every possible rational zero. $p \mid 4$ **gives** $p = \pm1, \pm2, \pm4$**;** $q \mid 2$ **gives** $q = \pm1, \pm2$**. The list (reduced, no repeats) is** $\pm1,\ \pm2,\ \pm4,\ \pm\frac{1}{2}$**.**

Answer: $\pm1,\ \pm2,\ \pm4,\ \pm\frac{1}{2}$

(b) Find all real zeros of f, showing the test and the division you used. **Test** $x = 2$**:** $f(2) = 16 - 4 - 16 + 4 = 0$**, so** $x = 2$ **is a zero. Synthetic division by 2 on** $2,\ -1,\ -8,\ 4$ **gives** $2,\ 3,\ -2,\ 0$**, so the depressed polynomial is** $2x^2 + 3x - 2 = (2x - 1)(x + 2)$**. Hence** $f(x) = (x - 2)(2x - 1)(x + 2)$ **and the zeros are** $x = 2,\ x = \frac{1}{2},\ x = -2$**. Note** $(2x - 1)$ **gives** $x = \frac{1}{2}$ **— not 2.**

Zeros: $2,\ \frac{1}{2},\ -2$

7. **Solve over the complex numbers.** $x^3 - 3x^2 + 4x - 12 = 0$

(a) Before solving: how many solutions must this equation have, counted with multiplicity, and which theorem says so?

Exactly 3, by the Fundamental Theorem of Algebra

(b) Solve. Give every solution. **Group:** $x^2(x-3) + 4(x-3) = (x-3)(x^2+4) = 0$. **So $x = 3$, or $x^2 = -4 \Rightarrow x = \pm 2i$. The solutions are $x = 3,\ 2i,\ -2i$ — three of them, as (a) promised.**

Solutions: $3,\ 2i,\ -2i$

(c) How many of your solutions are real, and how many x-intercepts does the graph of $y = x^3 - 3x^2 + 4x - 12$ therefore have? One real solution ($x = 3$), so exactly one x-intercept — a graph can never show the imaginary pair

8. **Build a polynomial from its zeros.** Write a polynomial function of *least degree* with *real* coefficients whose zeros are -1 and $2i$. Give your answer in factored form *and* in standard form. **Real coefficients force the conjugate $-2i$ to be a zero too, so there are three zeros and the least degree is 3. Factored: $f(x) = (x+1)(x-2i)(x+2i) = (x+1)(x^2+4)$. Standard: $f(x) = x^3 + x^2 + 4x + 4$.**

Factored: $(x+1)(x^2+4)$ **Standard:** $x^3 + x^2 + 4x + 4$

Teacher Note

Item 1: (e) is the Unit 3 habit-breaker — $(2, 0)$ is a *relative* max only, and an odd-degree polynomial has *no* absolute extrema; the range is all reals for the same reason. Accept interval or inequality notation throughout. Item 1(c): full credit requires the multiplicity *and* the cross/touch, since the touch at $x = 2$ is the only evidence of the squared factor. Item 3: no credit for $(x+3)(x^2-4)$ or $2x(x^2-9)$ — "completely" is the standard. Item 4(a): the placeholder $0x^2$ is the graded step; a quotient of degree 1 means it was skipped. Item 6(b): accept a graph-based choice of which candidate to test, but the division must appear. Item 7(b): watch for $\pm$ dropped on the square root, and for "no real solutions" written as the final answer — the question says *over* $\mathbb{C}$.

Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. **The graph plan.** Consider $f(x) = -2(x+2)(x-1)^2(x-3)$. Do *not* expand it. **(a)** State the degree and the leading coefficient, and explain how you got each one from the factored form. What end behavior do they force? **(b)** List the zeros with their multiplicities, and say whether the graph crosses or touches the x-axis at each. **(c)** Find the y-intercept. **(d)** Build a sign chart and state every interval on which $f(x) > 0$. **(e)** At most how many turning points can this graph have, and why?

(a) Degrees add: $1 + 2 + 1 = 4$, so the degree is 4. Leading coefficients multiply: $-2 \cdot 1 \cdot 1 \cdot 1 = -2$. An even degree with a negative leading coefficient sends *both* arms down: $f(x) \to -\infty$ at each end.

(b) $x = -2$ (multiplicity 1, crosses); $x = 1$ (multiplicity 2, touches); $x = 3$ (multiplicity 1, crosses).

(c) $f(0) = -2(2)(-1)^2(-3) = 12$, so the y-intercept is $(0, 12)$.

(d) Test one value in each interval, keeping only signs: $f(-3) < 0$, $f(0) = 12 > 0$, $f(2) = 8 > 0$, $f(4) < 0$. So $f(x) > 0$ on $(-2, 1) \cup (1, 3)$ — that is, for $-2 < x < 3$ except $x = 1$ — and $f(x) < 0$ on $(-\infty, -2) \cup (3, \infty)$. The sign does *not* flip at $x = 1$; that is the fingerprint of the even multiplicity.

(e) At most $4 - 1 = 3$ turning points, since a degree-n polynomial turns at most $n - 1$ times.

2. **Modeling.** An open-top box is made from a 12 in by 6 in sheet of cardboard by cutting a square of side x from each corner and folding up the sides. **(a)** Write the volume $V(x)$ in factored form, then expand it into standard form. **(b)** State the feasible domain of the model and justify it from the sheet's dimensions. **(c)** Find $V(1)$ and state its units. **(d)** Check that $x = 6$ makes $V(x) = 0$. Explain why 6 is a genuine zero of the polynomial but not a possible cut length.

(a) Cutting x from both ends of each side leaves a base of $(12 - 2x)$ by $(6 - 2x)$ and a height of x, so $V(x) = x(12 - 2x)(6 - 2x)$. Expanding: $(12 - 2x)(6 - 2x) = 4x^2 - 36x + 72$, so $V(x) = 4x^3 - 36x^2 + 72x$.

(b) Every dimension must be positive: $x > 0$, $12 - 2x > 0$ (so $x < 6$), and $6 - 2x > 0$ (so $x < 3$). The tightest of these wins, so the feasible domain is $0 < x < 3$ — the short side is what limits the cut.

(c) $V(1) = 1(10)(4) = 40$ cubic inches (in^3). Check against standard form: $4 - 36 + 72 = 40$ ✓

(d) $V(6) = 6(12 - 12)(6 - 12) = 6 \cdot 0 \cdot (-6) = 0$, so 6 really is a zero of the polynomial. But a 6-inch cut from a 6-inch side removes the entire width, and $6 - 2(6) = -6$ is a negative length — there is no box at all. The polynomial does not know about cardboard; only the feasible domain $0 < x < 3$ describes a real box.

Teacher Note

Scoring Part D (6 pts each). D1: 1 pt (a) degree 4 *and* lead -2 with both arms down; 2 pts (b) all three zeros with multiplicities and cross/touch; 1 pt (c) $(0, 12)$; 1 pt (d) $(-2, 1) \cup (1, 3)$ with the no-flip at $x = 1$ noted; 1 pt (e) 3, justified by $n - 1$. Give no credit in (a) for "degree 3" from counting factors — the exponents must be added. D2: 2 pts (a) factored *and* expanded; 2 pts (b) $0 < x < 3$ justified by $6 - 2x > 0$ (a domain of $0 < x < 6$ earns 1); 1 pt (c) 40 in^3 with cubic units; 1 pt (d) $x = 6$ recognized as a real zero of V but rejected because the width $6 - 2x$ would be negative. Do not accept "because it is too big" without the dimension.